

Estimating spatial firm productivity spillovers using UK microdata

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Dissertation submitted in part-fulfilment of the MSc in Economics

University College London (UCL)

September 2026

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DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistative role (UCL Category 2) for this submission. I used Gemini 3.10 Pro (?). This is limited to the roles of: supporting the particular process of reviewing and refactoring code (data analysis), as a support tutor (model and empirical approach).

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Date: 2nd September, 2026

CLASSIFICATION

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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ABSTRACT

~300 word summary of your research topic, methodology and findings

ACKNOWLEDGEMENTS

- Lars Nesheim
- John Muellbauer ⁴
- Przemek Karpisz ⁴

Word count:

⁴University of Oxford

1. INTRODUCTION

2. LITERATURE

My investigation is rooted in Marshall (1920)'s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers. There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman also labels all these channels as 'centripetal' to an urban core; there are likewise 'centrifugal' factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have "increasing strength" attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently (Bloom, Schankerman and Van Reenen, 2013).

Knowledge spillovers have been a well-evidence factor of within-firm managerial-derived productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

3. DATA

3.1. FAME

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular, number of employees and balance sheet data from which analysis-relevant aggregates can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey⁵.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim for (?). and reused for this project. Raw data is provided on a series of a financial balance sheet indicators, including characteristics that are fixed for each firm, bucketed into *a1_ID* and *a5_misc* indicator tables, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

3.2. INGESTION AND CLEANING

(...)

All data is deflated to real terms, using 2024 as the base year. I used the GDP deflator for all balance sheet variables, except for *renumeration_employees* and its composites for which I used CPIH, as these are income-related aggregates with real spending power best captured by household purchasing baskets⁶.

All balance sheet data are in thousands ('000s) of GBP (£). As we estimate log relationships exclusively, this has no effect on parameter interpretation.

3.3. LOCATION DATA

3.4. DERIVING PANEL AGGREGATES

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame's *fixed_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when

⁵?

⁶?

filtering for positive values Table (??), which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE I
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratios
<i>tangibles</i>		3.53
<i>fixed_total</i>	tangibles + intangibles + inv._other	6.00
<i>total_assets</i>	fixed_total + current_assets	11.56

GVA is calculated as the sum of *ebitda* and *renumeration_employees*. *renumeration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via the sum of *profit_loss_pretax*, *interest_paid*, and *depreciation*, providing an alternative measure of GVA. We can expect the difference between these two to be accounting differences⁷ and additionally note that the alternate GVA measure also reduces panel observations from 1,085,520 to 618,588. For these reasons, I do not report TFP calculations with this second GVA measure which are regardless near-identical, though I resultingly refer to GVA throughout this paper as *gva_1*.

3.5. FIRM DISTANCE MEASURES

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. This relationship is irreflexive. For groups it is transitive: if firms A and B share a spatial group, such as a travel-to-work area (*ttwa*), first-half of a UK postcode (*pc4*), or full postcode (*pc8*), then a firm C that is a peer of firm B must also be of firm A. For continuous distance measures, this is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are not shared with A. Every firm is then its own overlapping group, or equivalently, a sparse, rather than block diagonal, firm-to-firm matrix⁸.

(Cleaning address data).

Fame only reports each company’s last-known registered and trading address, at the time of compilation of publication of its annual vintage(cite). As a result, final location data is only available at the firm level. We can interpret each firm as being static over the entire 20-year sample, with no movers. This seems very unlikely (expand). Instead, we must interpret the constructing address, and therefore firm-to-firm distance measures as a signal of firm pairs’ true distance polluted by movers. This has 2 econometric consequences.

Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, their firm location decision is endogenous to the subsequent productivity spillovers they experience and exert, for reasons discussed in Section ?? on New Economic Geography. (More on firm FEs) There is no solution to this for moves occurring out-of-sample, for instance, just before 2009.

⁷An OLS regression between the two, discarding 5% outliers, has a coefficient of 0.91 with a 0.000 p-value. This means *gva2* is slightly lower than *gva1*, with that impact mostly driven by top-end outliers.

⁸I do actually impose a block diagonal structure for reasons of computation as explained subsequently

Secondly, I acknowledge some attenuation to the productivity spillover estimate Peer firms who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location (More on direction of bias/positive selection?)

I first consider a model where each target has peers which share a spatial identifier, meaning they belong to the same travel-to-work-area (*ttwa*) using the 2011 definitions ⁹, the first half of a UK postcode *pc4*, and the full UK postcode *pc8*.

(Discuss and illustrate different forms of omega, with illustrative block diagonal matrices)

⁹(Review ONS bulletin on TTWAs data source and describe clearly)

4. STATIC MODEL

I use a stylised 3-factor production model throughout this analysis. Y , A , K , and L are each firm-year observation's structural GVA, TFP, labour, and capital as usual. I use lower-case variables to indicate logged terms in reduced form and bolded variables to indicate stacked vectors.

Log TFP, denoted by $z_{i,t}$, is the reduced-form residual after controlling for the two classical inputs, which are assumed to be exogenous and exhibit unit factor substitutability in this static model. Under this specification, the residual is equivalent to the sum of a constant across all observations, firm and time fixed effects, and most crucially, the weighted sum of all peer log TFP $\sum_{j \neq i} w_{ij} z_{j,t}$, with a structural δ parameter to estimate.

$$Y_{i,t} = A_{i,t} \cdot K_{i,t}^{\beta_1} \cdot L_{i,t}^{\beta_2} \quad (1)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (2)$$

$$z_{i,t} = y_{i,t} - \alpha l_{i,t} - \beta k_{i,t} = \ln(A_{i,t}) \quad (3)$$

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \delta \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (4)$$

Theory suggests no restrictions on β_1 , β_2 , and δ collectively should be imposed, as δ is an externality. (Lucas discusses this)

The reduced-form equation ?? is linear in the weights w_{ij} applied to peer log TFP and its coefficient. This permits me to estimate structural δ through either the full GVA specification in one stage, using the log of *gva1* as the dependent variable, or in two stages, calculating $z_{i,t}$ as the 2-factor residual and then regressing $z_{i,t}$ however desired in reduced form. This is convenient for a number of the static models below.

4.1. 2-FACTOR TFP ESTIMATION

For convenience, I initially conduct the 2-factor estimation deriving the residual z_{ijt} to decompose further in a second stage. This step is additionally helpful for validating the relationships about the derived aggregates from my data.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + u_{i,t} \quad (5)$$

$$u_{i,t} = \delta \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (6)$$

I conduct the estimation using the log-transformed variables. I must include firm and time fixed effects and a constant if equation (??) is the true specification, otherwise the static panel OLS will inaccurately attribute firm and time variation to my classical factor inputs (explain more clearly). As outlined above, this analysis assumes factor input exogeneity.

The parameter coefficients give a labour income share exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour*

TABLE II
Baseline TFP calculation from 2-factor production

	(1)	(2)	(3)	(4)	(5)
	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁
constant	3.201 (0.032)***	4.840 (0.019)***	3.223 (0.027)***	1.994 (0.010)***	1.996 (0.010)***
ln(<i>employees</i>)	0.621 (0.005)***	0.717 (0.005)***	0.618 (0.004)***	0.559 (0.002)***	0.559 (0.002)***
ln(<i>fixed_total</i>)		0.058 (0.001)***			
ln(<i>total_assets</i>)	0.265 (0.004)***		0.264 (0.004)***	0.422 (0.002)***	0.422 (0.002)***
<i>i</i> fixed effects	✓	✓	✓		
<i>t</i> fixed effects	✓	✓		✓	
Observations	1,081,520	1,042,854	1,081,520	1,081,520	1,081,520
R ²	0.3701	0.3226	0.3861	0.8345	0.8346

All parameters reported with standard errors underneath. Significant estimates against an $H_0 : \beta = 0$ are indicated with $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***).

share which is constructed from a similar sample to the coverage in Fame¹⁰.

Surprisingly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income, or explanatory power of capital on variation in GVA. I leave it for future analysis why this dataset yields this phenomenon and for the rest of the analysis, I use *total_assets* as the primary capital measure.

As a demonstration, I include specifications which omit fixed effects. The capital share of income jumps on a specification with only time fixed effects, which is to be expected (explain).

TFP is the residual on GVA after extracting the proportion that can be accounted by capital and labour inputs. Equivalently, it is the sum of fixed effects in this specification with the error term, which means we should expect it to be exogenous to capital and labour inputs, but correlated to both firm effects and time trends. By extracting this regression residual, I have a TFP measure for all of my 1,081,520 panel observations.

4.2. STATIC PANEL IDENTIFICATION

Models in this section are consistent only if the regressors are plausibly exogenous from $\varepsilon_{i,t}$. Following standard procedure for a wide panel regression, meaning many firm observations and far fewer time periods, I transform all variables to first differences.

¹⁰?

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \delta \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t}$$

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \delta \sum_{j \neq i} w_{ij} \cdot (\alpha_i + \gamma_t + \beta_0 + \varepsilon_{j,t}) + \varepsilon_{i,t} \quad (7)$$

$$\Delta z_{i,t} = \Delta \gamma_t + \delta \sum_{j \neq i} w_{ij} \cdot (\Delta \gamma_t + \Delta \varepsilon_{j,t}) + \Delta \varepsilon_{i,t} \quad (8)$$

For these models, I make the strong assumption of no structural, cross-sectional correlation between error terms. This means that for any productivity shock $\varepsilon_{j,t}$ that hits a peer firm j , if it is correlated with a productivity shock $\varepsilon_{i,t}$ occurring in the target firm i within the same period t , it must be attributable to the peer effect coefficient δ .

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] = 0 \quad (9)$$

This is quite a strong assumption. It precludes any contemporaneous, correlated independent TFP shocks, which might occur on both the global and local level. (...)

4.3. LMM GROUPED PEERS

$$\Delta z_{i,t} = \Delta \gamma_t + \delta \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta \varepsilon_{i,t}$$

$$\Delta \mathbf{z}_t = \Delta \gamma_t \mathbf{1} + \delta \mathbf{W} \Delta \mathbf{z}_t + \Delta \varepsilon_t$$

$$= (\mathbb{I} - \delta \mathbf{W})^{-1} (\Delta \gamma_t \mathbf{1} + \Delta \varepsilon_t)$$

Section ?? discusses how firms have been labelled to symmetric, transitive, non-overlapping groups by *pc8* (smallest), *pc4*, and *ttwa* (largest). I conduct a linear regression using the averages of peers belonging to these groups, leaving out the target firm. This is known as the 'Linear-in-Means (LMM)' Model and is common to the peer effects literature.

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \sum_{g=1}^3 \beta_g \mathbb{E}[z_{jtg} | g] + \varepsilon_{it} \quad (10)$$

$$\mathbf{z}_t = \delta \mathbf{W} \mathbf{z}_t$$

$$w_{ij} \in \mathbf{W} = \begin{cases} 1, & \text{if } (i, j) \in g \\ 0, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

It is crucial that I have multiple, non-overlapping groups in a single model, where my target firm z_{it} is a member of all groups simultaneously, but peers in one group are excluded from all others.

The non-overlapping property prevents multicollinearity. If I were to use nested, overlapping geographies, namely keeping all firms within the *ttwa* area in that group average,

including the ones I have just counted in the $pc4$ and $itxpc8$ groups, then my group averages mechanically align: $z_{jt,pc8}$ is embedded within $z_{jt,pc4}$, and both are within $z_{jt,ttwa}$, making β_g parameter estimates unstable.

The single model approach prevents omitted variable bias (OVB). $\mathbb{E}[z_{jt,ttwa}|g] \propto \mathbb{E}[z_{jt,pc8}|g]$, so running three models where the former is just my single regressor means the latent $\beta_{pc8} \in u_{it}$ is correlated with both z_{itg} and z_{jtg} . We could reasonably argue that the inner rings are causal, so this causal factor excluded from the regression influences both and creates heavily upward bias on my outer-ring estimations. Indeed, this critique extends within further inner rings. If I think the causal unit is not peer firms within the $pc8$ group at all, but a further hyper-local effect - perhaps those on the same street or same building, then this latent factor will correlate with both my outcome and regressor and drive the results. However, given the small number of peers contained within an $pc8$ unit and the limits of data availability to identify firms at more local granularities, I proceed assuming the effect from $pc8$ peers is homogenous.

Finally, the approach using multiple groups as regressors at once gives identification by breaking transitivity. If I were to run the same regression with merely one group, say $pc8$, then my regression would suffer the reflection problem outlined in (?). As I am estimating the expected value of z_{it} and z_{jt} from the same sample, they cannot be uniquely identified in reduced form.

Reflection problem with a single regressor, ι denotes a column vector of 1s

(review: maths is a bit tricky here)

$$\begin{aligned}\mathbb{E}[z_{itg}|g] &= \alpha_i + \gamma_t + \beta_0 + \beta_1 \mathbb{E}[z_{jtg}|g] \\ &= (\alpha_i + \gamma_t + \beta_0)\iota + \beta_1 \mathbf{\Omega}_t \mathbb{E}[\mathbf{z}_{tg}|g] \\ &= (\mathbb{I} - \beta_1 \mathbf{\Omega}_t)^{-1}(\alpha_i + \gamma_t + \beta_0)\iota\end{aligned}$$

This problem is mitigated with having 3 distinct groups to which z_{it} belongs. The results of the LMM estimation implementing equation ?? are below:

The parameter results suggest a strong correlation between the TFP of a target firm and its peers.

TABLE III
Static panel linear-in-means using spatial-group TFP averages

	(1) <i>tfp</i>	(2) <i>tfp</i>	(3) <i>tfp</i>
constant	2.075 (0.044)***	0.310 (0.038)***	1.202 (0.215)***
$\ln(\text{employees})$			0.184 (0.044)***
<i>peer_tfp_pc8</i>	0.280 (0.005)***	0.559 (0.004)***	0.280 (0.005)***
<i>peer_tfp_pc4_donut</i>	0.015 (0.005)***	0.142 (0.008)***	0.015 (0.005)***
<i>peer_tfp_ttwa_donut</i>	0.033 (0.013)***	0.204 (0.013)***	0.306 (0.071)***
<i>peer_tfp_pc8_x_employees</i>			-0.058 (0.015)***
<i>i</i> fixed effects	✓		✓
<i>t</i> fixed effects	✓	✓	✓
Observations	764,537	764,537	764,537
R ²	0.0388	0.1792	0.0390

4.4. STATIC DISTANCE DECAY

To improve on the LMM specification, I make a modification where every peer firm is weighted by its influence on the target by the reciprocal of the continuous distance metric calculated in ?? $w_{ijt} = f(d)$. This is stylistic of a declining peer influence further away, representing a (lower frequency of social collisions, barriers created via transport and time costs, and an increased difficulty of observation of management practices).

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \beta_1 \sum_{j \neq i} w_{ijt} z_{jt} + \varepsilon_{it} \quad (12)$$

$$\mathbf{z}_t = (\alpha_i + \gamma_t + \beta_0)\mathbf{1} + \beta_1 \sum_{j \neq i} \Omega_t \mathbf{z}_t + \varepsilon_t \quad (13)$$

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2} & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (14)$$

This represents a likely more accurate declining effect of productivity spillover, rather than the special-case group mean where every firm in an enclosed area, for example, the target's postcode, has an equal influence on the target's TFP. In addition, it allows for better identification following ?. (explain Bramouille2009, patterns of) Ω_t

In essence, the LMM specification only contains 3 non-overlapping groups from its location tags. The distance-decayed specification contains as many non-overlapping groups as firms.

This distance-decay specification revises down the estimate from peers to a an elasticity

TABLE IV
Static panel using continuous distance-decay weighted
TFP average of peers

	(1)	(2)	(3)
	<i>tfp</i>	<i>tfp</i>	<i>tfp</i>
constant	2.479 (0.029)***	2.586 (0.029)***	2.417 (0.032)***
$\ln(\textit{employees})$	-0.009 (0.004)**	-0.009 (0.004)**	-0.009 (0.004)**
$\ln(\textit{nb_peers})$	0.013 (0.004)***	0.012 (0.004)***	0.016 (0.004)***
<i>tfp_wav</i> ₁	0.168 (0.003)***		
<i>tfp_wav</i> ₂		0.135 (0.003)***	
<i>tfp_wav</i> ₃			0.184 (0.005)***
<i>i</i> fixed effects	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓
Observations	1,067,013	1,067,013	1,067,013
R ²	0.0101	0.0095	0.0040

of 0.168, lower than the results given by the LMM specification. This is likely due to (...) In addition, I introduce controls (...)

5. DYNAMIC MODEL

I employ a dynamic model for additional identification to help eliminate the static model $\mathbb{E}[\varepsilon_{i,t}\varepsilon_{j,t}] = 0$ strong assumption.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (15)$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta \sum_{j \neq i} w_{i,j,t} \cdot (\alpha_j + \gamma_t + \beta_0 + \varepsilon_{j,t}) + \varepsilon_{i,t} \quad (16)$$

To support this argument, I introduce a dynamic panel specification, such that lagged realisations of the dependent variable are used as additional regressors. I can implement these lags exploiting the rich time series observations available in Fame.

Autoregressive lags help control for reverse causality: if $y_{i,t-k}$ is causal for $z_{j,t}$, or feasibly $z_{j,t-1}$ and $\mathbb{E}[z_{j,t-1}, z_{j,t}] \neq 0$, then the presence of $y_{i,t-k}$ as a control helps eliminate that bias. Similarly, theory offers us no clear view on where there could be simultaneity, where variation in $\Delta y_{i,t}$ is driven causally by $\Delta y_{i,t-1}$ and that change in $\Delta y_{i,t}$ in turn drives $\Delta z_{j,t}$. The autoregressive regressor helps control for that. Additionally, the results below suggest the autoregressive parameter coefficient is low, with a log elasticity of 0.005 (significant at the 1% level) in all specifications, suggesting any concerns about reverse causality or simultaneity driven by an AR process are likely minimal. The use of this dynamic regressor improves the plausible exogeneity of $z_{i,j,t}$.

The autoregressive specification requires a return to the one-stage production function with an outcome of GVA, y_{it} . I continue to use the scalar weighted distance TFP metric as the primary regressor of interest $z_{j,t}$. While introducing dynamic lags of

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_q \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (17)$$

I list an AR(p,q) specification above, however, my results show that AR(1), or $p = 1$, is the true structure. Appendix I, which increases the order using a nested model approach, shows that for any specification order above an AR(1) coefficients become insignificant suggesting multicollinearity (F-test on AR(2)). I also confirm this via information criteria tests, (which show) that lags beyond $t - 2$ provide no new information for the variation on $\ln(y_{i,t})$. Noting that the regression coefficient on the $\ln(gva_1)_{t-1}$ is significant at the 1% level in my results, I can strongly infer that an AR(1) is the true structure.

((IC test results))

5.1. INSTRUMENTAL IDENTIFICATION UNDER NO ERROR AUTOCORRELATION

Firstly, I assume that dynamic specification (??) is well-specified, crucially that means the error term $\varepsilon_{i,t}$ exhibits no autocorrelation, though as discussed I allow for moderate cross-sectional correlation across firms.

$$\mathbb{E}[\epsilon_{i,t}\epsilon_{i,t-1}] = 0, \quad \forall t, t-1 \quad (18)$$

$$\mathbb{E}[\epsilon_{i,t}\epsilon_{j,t}] \neq 0, \quad \exists i, j \quad (19)$$

As usual, a panel setting requires a transformation to strip out firm fixed effects. I continue to use first differences of my log aggregates, however, even under the fairly strong assumption (??), the first-order autoregressor, $\Delta \ln(y_{it})$ is endogenous.

$$\begin{aligned} \Delta y_{i,t} &= \Delta \gamma_t + \sum_{k=1}^p \phi_k \Delta y_{i,t-k} + \theta_1 \Delta k_{i,t} + \theta_2 \Delta l_{i,t} + \sum_{k=1}^q \delta_q \sum_{j \neq i} \Delta(w_{i,j,t} z_{j,t}) + \Delta \varepsilon_{i,t} \quad (20) \\ \Delta \varepsilon_{i,t} &= \varepsilon_{i,t} - \varepsilon_{i,t-1} \\ \Delta y_{i,t-1} &= \dots + \Delta \varepsilon_{i,t-1} \\ &= \dots + \varepsilon_{i,t-1} - \varepsilon_{i,t-2} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] \neq 0 \\ \Delta y_{i,t-2} &= \dots + \varepsilon_{i,t-2} - \varepsilon_{i,t-3} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] = 0 \end{aligned}$$

The treatment of the other standard factors, $\Delta k_{i,t}$ and $\Delta l_{i,t}$

$$K_{i,t} = \lambda I_{i,t} + (1 - \lambda) I_{i,t-1} + d K_{i,t-1} \quad (21)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* | \Omega_t] \quad (22)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* | \Omega_t] \quad (23)$$

$$\epsilon_t \in \Omega_t$$

$$\implies K_{i,t} \propto \epsilon_t, \epsilon_{t-1}$$

$$\implies L_{i,t} \propto \epsilon_t$$

$$\Delta k_{i,t} = k_{i,t} - k_{i,t-1}$$

$$\implies \mathbb{E}[\Delta k_{i,t}, \Delta \epsilon_t] \neq 0$$

$$\implies \mathbb{E}[\Delta k_{i,t-1}, \Delta \epsilon_t] \neq 0$$

$$\implies \mathbb{E}[\Delta k_{i,t-2}, \Delta \epsilon_t] = 0$$

This is confirmed by the Anderson-Rubin (AR(1)) test in my results, which rejects the null hypothesis at the 1% level for all models, suggesting that first-order lags of log GVA are a weak instrument (Why? Shouldn't they fail exclusion restriction, not instrument correlation). In addition, I assume that capital, $k_{i,t}$ is pre-determined, as investment is chosen one period ahead, in line with standard Perpetual Inventory Method approaches¹¹ (Review

¹¹Noting that this is ordinarily done on a quarterly basis, however, due to data limitations we adopt the same

predetermined material. Additionally, explain the logic of instrumenting for capital)

To overcome this endogeneity, I introduce deeper autoregressive lags as instrumental variables, as proposed by (?). I use 3 autoregressive instrument lags starting from $t - 2$: $\ln(y_{i,t-2})$, $\ln(y_{i,t-3})$, $\ln(y_{i,t-p-3})$ for log GVA, and 2 lags for capital $\ln(k_{i,t-p-1})$, $\ln(k_{t-p-2})$. An Anderson-Rubin (AR(2)) test shows that deep lags are a valid instrument for log GVA. The test applied to the second lag of $\ln(gva_1)$ fails to reject the null hypothesis of for any specification, so I assume it is a valid instrument.

Additionally, using multiple deep lags allows for overidentification: in this basic AR(1) specification, I have $k = 4$ regressors and $l = 7$ instruments. I could increase this. Fame is rich on the time dimension, with up to 19 lags at most for a firm, and a median of 6 lags, so I could safely provide more lags of regressors as instruments. Further lags is likely to have a decreasing marginal effect on explanatory power and additionally ((...))

(Discuss introducing a new weighting method using past firm similarity.)

I estimate the equation fulfilling the vector of moment conditions, (??) via GMM.

$$\mathbb{E} \left[\Delta \varepsilon_{it} \cdot \begin{pmatrix} \Delta \ln(y_{i,t-2}) \\ \Delta \ln(y_{i,t-3}) \\ \Delta \ln(y_{i,t-4}) \\ \Delta \ln(k_{i,t-2}) \\ \Delta \ln(k_{i,t-3}) \\ \Delta \Sigma_j w_{ijt} z_{jt} \end{pmatrix} \right] = 0 \quad (24)$$

As my regressors are over-identified, I use 2-step GMM to improve efficiency. (I should introduce a control (basically a diff-in-diff?) of peers from the same industry across the country (outside the spatial bubble).)

Finally, a Sargan-Hansen J-test is implemented to check if these instruments are over-identified, meaning the moment conditions cannot be fulfilled with 95% confidence. In all models, the J-test fails to reject the null hypothesis, meaning the model's moment conditions are jointly correct.

5.2. INSTRUMENTAL IDENTIFICATION UNDER ERROR AUTOCORRELATION

I appear to have strong identification given assumptions (??) and (??). The instruments above exhibit strong correlation under Anderson-Rubin tests and exclusion restriction has been derived. However, there exists a niche case of diagonally-correlated errors, namely despite (??) and (??), I ought to consider the following:

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t-k}] \neq 0, \quad \exists i, j, k \quad (25)$$

This is not as implausible as it may appear due to this analysis' peer effects setting. There may be some local omitted shock ν_t , that impacts peer firm A's TFP via ε_t and impacts target firm B's TFP after a lag via ε_{t-k} . This could be a price shock with random adjustment (or)

Additionally, relaxing (??) to allow for weak error autocorrelation between limited $\varepsilon_{i,t}$

thing here even though Fame provides observations annually.

TABLE V
Dynamic panel: all instrumented variables

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	0.653 (0.345)	0.031 (0.215)	-0.656 (0.970)
$\ln(gva_1)_{t-1}$	0.006 (0.001)***	0.003 (0.001)***	0.004 (0.002)
tfp_wav_{1t-1}			-0.017 (0.012)
$\ln(employees)$	0.523 (0.051)***	0.655 (0.027)***	0.599 (0.075)***
$\ln(total_assets)$	0.310 (0.030)***	0.298 (0.030)***	0.283 (0.036)***
tfp_wav_1	0.832 (0.087)***	0.887 (0.081)***	1.251 (0.303)***
Hansen Test	✓ (0.049)**	✓ (0.015)**	✗ (0.941)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	✗ (0.119)	✗ (0.879)	✗ (0.117)
Instruments	35	35	52
i fixed effects	✓	✓	✓
t fixed effects	✓	✓	✓
Observations	713,339	713,339	713,339

Model includes a constant as the baseline for t fixed effects.

TABLE VI
Dynamic panel using instrumented continuous
distance-decay weighted TFP

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	1.371 (0.134)***	1.577 (0.144)***	1.064 (0.103)***
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
$\ln(employees)$	0.696 (0.038)***	0.689 (0.037)***	0.709 (0.040)***
$\ln(total_assets)$	0.259 (0.041)***	0.266 (0.041)***	0.247 (0.044)***
tfp_wav_1	0.503 (0.028)***		
tfp_wav_2		0.423 (0.023)***	
tfp_wav_3			0.618 (0.042)***
Hansen Test	X (0.417)	X (0.459)	X (0.246)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.746)	X (0.600)	X (0.945)
i fixed effects	✓	✓	✓
t fixed effects	✓	✓	✓
Observations	713,339	713,339	713,339

Model includes a constant as the baseline for t fixed effects.

and $\varepsilon_{i,t-k}$ for limited k proves a similar problem.

(Firstly, I test for error autocorrelation.)

5.3. PANEL VECTOR AUTOREGRESSION

Another mechanism offered by this dynamic setting is to extremely flexibly introduce additional lags of my peer effect term, $\sum_{j \neq i} w_{i,j,t} z_{j,t}$.

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_q \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (26)$$

The model now has the form of a panel VAR(p,q) with internal instruments, where p refers to the autoregressive lags on the dependent variable $y_{i,t}$ and q refers to the regressor lags on $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$ chosen by my specification.

Following above reasoning, lags of $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$ of a higher order than 2 are exogenous to my reduced-form specification. However,

5.4. ALTERNATIVE WEIGHTS ON PEER TFP

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_q \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (27)$$

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APPENDIX 1

TABLE VII
Dynamic panel: higher-order autoregressive lags

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$
constant	1.371 (0.134)***	1.446 (0.843)	0.208 (2.578)	0.459 (2.844)
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.019 (0.104)	0.128 (0.295)	-0.020 (0.256)
$\ln(gva_1)_{t-2}$		0.002 (0.003)	0.004 (0.101)	0.094 (0.300)
$\ln(gva_1)_{t-3}$			-0.001 (0.006)	0.036 (0.162)
$\ln(gva_1)_{t-4}$				-0.003 (0.004)
$\ln(employees)$	0.696 (0.038)***	0.762 (0.097)***	0.736 (0.149)***	0.659 (0.323)*
$\ln(total_assets)$	0.259 (0.041)***	0.190 (0.090)*	0.248 (0.121)*	0.291 (0.305)
tfp_wav_1	0.503 (0.028)***	0.548 (0.059)***	0.508 (0.076)***	0.466 (0.184)*
Hansen Test	X (0.417)	X (0.922)	X (0.881)	X (0.826)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	X (0.140)	X (0.212)
AR(2) Test	X (0.746)	X (0.925)	X (0.802)	X (0.668)
i fixed effects	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185

TABLE VIII
Dynamic panel: higher-order regressor lags 1-5

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$
constant	1.124 (0.113)***	0.956 (0.101)***	0.743 (0.093)***	0.584 (0.091)***	0.543 (0.094)***
$\ln(gva_1)_{t-1}$	0.002 (0.001)*	0.003 (0.001)***	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
tfp_wav_{1t-1}	0.066 (0.006)***	0.077 (0.007)***	0.069 (0.007)***	0.062 (0.007)***	0.061 (0.008)***
tfp_wav_{1t-2}		0.052 (0.005)***	0.055 (0.006)***	0.050 (0.007)***	0.051 (0.007)***
tfp_wav_{1t-3}			0.043 (0.005)***	0.048 (0.006)***	0.049 (0.006)***
tfp_wav_{1t-4}				0.030 (0.005)***	0.038 (0.006)***
tfp_wav_{1t-5}					0.028 (0.005)***
$\ln(employees)$	0.686 (0.036)***	0.695 (0.038)***	0.670 (0.040)***	0.644 (0.044)***	0.665 (0.052)***
$\ln(total_assets)$	0.270 (0.040)***	0.260 (0.042)***	0.287 (0.045)***	0.317 (0.049)***	0.294 (0.057)***
tfp_wav_1	0.508 (0.028)***	0.514 (0.029)***	0.492 (0.030)***	0.469 (0.033)***	0.480 (0.038)***
Hansen Test	X (0.492)	X (0.968)	X (0.453)	X (0.084)*	X (0.080)*
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.279)	X (0.612)	X (0.801)	X (0.854)	X (0.729)
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185	365,908

TABLE IX
Dynamic panel: higher-order regressor lags 6-10

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$
constant	0.421 (0.082)***	0.388 (0.081)***	0.270 (0.077)***	0.199 (0.083)*	0.116 (0.093)
$\ln(gva_1)_{t-1}$	0.004 (0.001)***	0.004 (0.001)**	0.003 (0.001)*	0.002 (0.002)	0.002 (0.002)
tfp_wav_{1t-1}	0.055 (0.008)***	0.056 (0.008)***	0.047 (0.008)***	0.042 (0.009)***	0.036 (0.011)**
tfp_wav_{1t-2}	0.045 (0.007)***	0.045 (0.007)***	0.040 (0.007)***	0.037 (0.007)***	0.035 (0.009)***
tfp_wav_{1t-3}	0.046 (0.006)***	0.046 (0.006)***	0.039 (0.006)***	0.033 (0.007)***	0.029 (0.008)***
tfp_wav_{1t-4}	0.034 (0.006)***	0.031 (0.006)***	0.024 (0.006)***	0.019 (0.007)**	0.017 (0.008)*
tfp_wav_{1t-5}	0.029 (0.005)***	0.030 (0.006)***	0.023 (0.006)***	0.019 (0.006)**	0.017 (0.006)**
tfp_wav_{1t-6}	0.018 (0.004)***	0.024 (0.005)***	0.021 (0.005)***	0.019 (0.005)***	0.017 (0.006)**
tfp_wav_{1t-7}		0.019 (0.005)***	0.017 (0.005)***	0.015 (0.006)**	0.011 (0.007)
tfp_wav_{1t-8}			0.013 (0.004)**	0.016 (0.005)**	0.017 (0.006)**
tfp_wav_{1t-9}				0.010 (0.005)*	0.013 (0.005)*
$\ln(employees)$	0.628 (0.049)***	0.636 (0.053)***	0.582 (0.055)***	0.548 (0.061)***	0.523 (0.073)***
tfp_wav_{1t-10}					0.016 (0.006)**
$\ln(total_assets)$	0.334 (0.054)***	0.325 (0.058)***	0.384 (0.061)***	0.421 (0.067)***	0.448 (0.080)***
tfp_wav_1	0.452 (0.036)***	0.457 (0.038)***	0.420 (0.040)***	0.397 (0.044)***	0.380 (0.052)***
Hansen Test	X (0.097)*	X (0.122)	X (0.450)	X (0.637)	X (0.435)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.924)	X (0.962)	X (0.568)	X (0.507)	X (0.450)
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	307,394	256,016	210,798	171,125	136,527

TABLE X
Dynamic panel: higher-order VAR(2,4) structures

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$	(6) $\ln(gva_1)$
constant	1.164 (0.573)*	0.498 (1.561)	0.398 (1.667)	1.494 (0.428)***	0.333 (1.469)	1.351 (0.307)***
$\ln(gva_1)_{t-1}$	-0.011 (0.117)	0.081 (0.280)	-0.028 (0.251)	-0.129 (0.081)	0.087 (0.276)	-0.125 (0.063)*
$\ln(gva_1)_{t-2}$	-0.000 (0.003)	0.006 (0.111)	0.076 (0.385)	0.004 (0.002)	-0.001 (0.111)	0.006 (0.002)**
$\ln(gva_1)_{t-3}$		-0.001 (0.005)	0.034 (0.156)		-0.002 (0.004)	
$\ln(gva_1)_{t-4}$			-0.002 (0.004)			
tfp_wav_1	0.551 (0.052)***	0.504 (0.078)***	0.475 (0.201)*	0.525 (0.049)***	0.488 (0.073)***	0.524 (0.050)***
tfp_wav_{1t-1}	0.099 (0.102)	0.005 (0.255)	0.082 (0.217)	0.196 (0.073)**	-0.005 (0.249)	0.189 (0.055)***
tfp_wav_{1t-2}	0.060 (0.011)***	0.036 (0.099)	-0.019 (0.342)	0.072 (0.012)***	0.046 (0.102)	0.070 (0.010)***
tfp_wav_{1t-3}				0.056 (0.009)***	0.039 (0.020)	0.065 (0.008)***
tfp_wav_{1t-4}						0.042 (0.006)***
$\ln(employees)$	0.737 (0.099)***	0.720 (0.146)***	0.643 (0.382)	0.650 (0.075)***	0.705 (0.148)***	0.657 (0.079)***
$\ln(total_assets)$	0.209 (0.088)*	0.253 (0.119)*	0.307 (0.367)	0.272 (0.074)***	0.271 (0.119)*	0.266 (0.079)***
Hansen Test	X (0.971)	X (0.855)	X (0.849)	X (0.999)	X (0.851)	X (0.232)
AR(1) Test	✓ (0.002)***	X (0.124)	X (0.163)	✓ (0.000)***	X (0.111)	✓ (0.000)***
AR(2) Test	X (0.889)	X (0.878)	X (0.775)	X (0.092)*	X (0.837)	✓ (0.030)**
i fixed effects	✓	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓	✓
Observations	603,058	511,396	433,185	511,396	511,396	433,185

APPENDIX 2

APPENDIX 3